

Designing Extended-Reality (XR) Experiences

Thursday 9:00 AM - 11:45 AM, [MIX 324](#)

Instructor	Instructor: Tejaswi Gowda, PhD Office Location: UGB 401B Email: tejaswi@asu.edu Telephone: 623-252-9347 Office hours: <i>Thursday 11:45-12:30 and by appointment</i> Preferred Contact Method: Email, Slack (ASU) Class Repo : https://github.com/tejaswigowda/game551-Fall2026 Slides: https://docs.google.com/presentation/d/1fqoHvb76n3efM5ZT3SmW6NNtjddM88fgsMW1lyNHrao/edit?slide=id.p#slide=id.p
Course Description	This course is a hands-on introduction to extended reality (XR) built on the open web as the common substrate. Students start with WebXR. No proprietary engine, no app-store gatekeeping, running on any desktop, headset or phone and progressively add the capabilities that define contemporary spatial computing: digital twins of real places, digital-physical co-design, on-device SLAM for world-anchored AR, motion capture, and large-scale immersive display (LED volumes). The through-line is the open web as a sovereign XR platform: everything students build is a URL they own, works offline, and does not depend on a store or a vendor. The course treats XR not as one device but as a pipeline – capture, compose, publish, inhabit and each unit maps to one stage of that arc, so the

	course material builds toward a single mental model rather than scattering across unrelated tools. Designed for graduate students in engineering, computing, and media technologies, this course prepares participants with the technical expertise and creative skills needed to capture, compose, publish, and inhabit spatial experiences on open, sovereign infrastructure.
Enrollment Requirements	<i>Programming experience preferred. GAME 220, GAME 294 or equivalent.</i>
Course Objectives	To equip students with the knowledge and hands-on skills to capture, compose, publish, and inhabit spatial experiences on open, sovereign web infrastructure, across digital twins, digital-physical co-design, world-anchored AR, motion capture, and immersive display.
Student Learning Outcomes	By the end of this course, students will be able to: • Build and deploy a WebXR experience that runs on headset and phone from a URL they own, and explain the principles of AR, VR, and XR. • Explain the XR pipeline – capture, compose, publish, inhabit and place any tool or technique within it. • Capture a real place or object into a 3D digital representation and publish it as a navigable digital twin, distinguishing experiential from measurement twins. • Implement world-anchored AR using device pose and SLAM, and reason correctly about coordinate frames (local, world, and global/GPS). • Build a digital–physical co-design loop: photograph a real object, reconstruct and edit it digitally, and return it to the physical world (e.g., 3D print). • Integrate motion-capture data into a real-time spatial experience, and describe how large-scale immersive display (LED volumes) differs from headset XR. • Collaborate effectively on projects that integrate both technical implementation and creative content design for XR. • Critically discuss XR's affordances, limits, and ethics – sovereignty, privacy of spatial data, accessibility, and who controls

	spatial platforms. • Produce a final project that demonstrates an end-to-end XR experience integrating the pipeline from capture through inhabiting, with engaging user interaction. These objectives prepare students to tackle real-world challenges in XR development across domains such as entertainment, cultural heritage, digital twinning, agriculture, and immersive media.
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Course Specifics

List of Primary and Secondary Materials Needed	This course can be accessed by both my.asu.edu and myasucourses.asu.edu; bookmark both in the event that one site is down. Additional Requirements This course requires the following technologies • Computer with Webcam, microphone, headset/earbuds, and speaker. • Web browsers (Chrome, Mozilla Firefox, or Safari) • Reliable broadband internet connection (DSL or cable). • Simple Text Editor (eg: TextMate on Mac, Notepad on Windows, VI on Unix) • GitHub account (Free account) • Minimal hardware requirement: Mac, Linux Box or Windows PC • AWS student access – provided • Other required software will be specified in class. Classroom Technology It is encouraged that you bring technology (cell phones, tablets and laptops) to class to help you take notes and do research, however please turn off cell phone ringers and do not use your phone to make personal calls in class or use any technology to use social media in class. Do not answer your phone in class. If you believe you are receiving an emergency call, please step outside to take it.
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	This course does not have a text-book. Appropriate reading material and tutorials will be assigned. Please follow along with the course on the class repo: https://github.com/tejaswigowda/game551-Fall2026 You will need a simple Text Editor (eg: Atom on Mac, Notepad++ on Windows, VI on Unix) and a GitHub account (free account).																				
Assignments	All assignments will be turned in on GitHub. Weekly builds are low-stakes and iterative; four projects: (1) a WebXR gallery, (2) a twin plus a digital–physical co-design artifact, (3) a world- or location-anchored AR piece, and (4) a presence or LED-volume piece are the substantial deliverables, and a final project integrates the full capture → compose → publish → inhabit pipeline. Details for each are introduced in class.																				
Grading	Assignments 1-6 are worth 10% of your grade and Assignments 7 and 8 are worth 20% of your grade. The final exam will be an opportunity for 10% extra credit. You will be given a one and a half hour task similar to one a prospective employer would give you to test your skill in coding. This brings the total amount of points in the class to 110 points and gives you the possibility of getting ten points extra credit.																				
Grade Scale	This course uses +/- grading <table border="1"> <tbody> <tr><td>98+</td><td>A+</td></tr> <tr><td>93-97</td><td>A</td></tr> <tr><td>90-92</td><td>A-</td></tr> <tr><td>88-89</td><td>B+</td></tr> <tr><td>83-87</td><td>B</td></tr> <tr><td>80-82</td><td>B-</td></tr> <tr><td>78-79</td><td>C+</td></tr> <tr><td>70-77</td><td>C</td></tr> <tr><td>60-69</td><td>D</td></tr> <tr><td>0-59</td><td>E</td></tr> </tbody> </table>	98+	A+	93-97	A	90-92	A-	88-89	B+	83-87	B	80-82	B-	78-79	C+	70-77	C	60-69	D	0-59	E
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Course Schedule

The instructor reserves the right to change portions of this syllabus (assignments, deadlines etc.) by verbal instructions during scheduled class time. The student is responsible for noting changes and acting accordingly. Grading and absence policies are not subject to change.

Date	Title	Topics Covered
Week 1	Substrate: WebXR & the Open Web	• The XR pipeline: capture, compose, publish, inhabit • Why WebXR: sovereign, cross-device, a URL you own • Deploy your first 'hello XR' scene to a URL
Week 2	three.js & the Scene	• Scene graph, camera, meshes, materials • Coordinate systems in 3D • Loading and placing glTF/GLB assets • Interactive 3D in the browser
Week 3	Into the Headset: WebXR Sessions	• WebXR device API; VR vs AR sessions • Controllers and interaction • Reference spaces (coordinate frames, part 1) • Deliverable: Project 1 a WebXR gallery • Deploy to headset and phone
Week 4	Digital Twins I: Capturing the Real	• Photogrammetry & single-image 3D reconstruction • What a twin is: twin vs model vs game • Experiential vs measurement twins

Week 5	Digital Twins II: Places & Time	• Capture an object into a WebXR scene • Twinning a space, not just an object • Twins that change over time vs static capture • Declarative scene manifests: what is where • Compose and publish a multi-object twin
Week 6	Digital–Physical Co-Design	• The loop: capture → edit → fabricate • Segment, retexture, clean a captured object • Deliverable: Project 2 : twin + co-design artifact (3D print)
Week 7	AR Foundations & Coordinate Frames	• Local vs world vs global frames • Marker/image-anchored AR • Build an image-anchored AR experience
Week 8	SLAM & World Anchoring	
Week 9	Global Anchoring: GPS & Control Points	• Location-based AR; why GPS alone isn't enough • Anchoring a local frame to global control points • Deliverable: Project 3 : a world/location-anchored AR piece

Week 10	Motion Capture I	• Mocap approaches; the Mesquite Mocap suit • Skeletons, retargeting, driving an avatar • Capture a motion; drive a character in WebXR
Week 11	Motion Capture II: Real-Time Presence	• Streaming mocap into a live experience; latency • The body as input; a real-time mocap interaction
Week 12	LED Volumes & Large-Scale Immersion	
Week 13	Ethics, Sovereignty & the Spatial Commons	
Week 14	Final Showcase	

Attendance Policy

<The instructor's general policy AND university policy on absences due to religious observance and university sanctioned activities. (Your general absence policy, first paragraph, will vary from the below)>

Instructor Attendance and participation policy: For example, for the duration of the class period is mandatory. If you have more than 3 absences (unexcused), your final grade will be lowered 1/3 grade for each subsequent absence (i.e. B to B-). You should notify me by email prior to absence if possible and provide doctor's note where applicable. Repeated tardiness and leaving class early will be recorded, and as a result, your final grade will be lowered. It is the student's responsibility to keep track of their absences.

	Excused absences related to religious observances/practices in accord with ACD 304–04, “Accommodation for Religious Practices.” Students may be excused for the observance of religious holidays outlined at: https://eoss.asu.edu/cora/holidays. Students should notify the instructor at the beginning of the semester about the need to be absent from class due to religious observances. Students will be responsible for materials covered during their absence and should consult with the instructor to arrange reasonable accommodation for missed exams or other required assignments. Excused absences related to university sanctioned activities in accord with ACD 304–02, “Missed Classes Due to University-Sanctioned Activities.” Students required to miss classes due to university sanctioned activities will not be counted absent. However, absence from class or examinations due to university-sanctioned activities does not relieve students from responsibility for any part of the course work required during the period of the absence. Students should inform the instructor early in the semester of upcoming scheduled absences and immediately upon learning of unscheduled required class absences. Reasonable accommodation to make up missed exams or other required assignments will be made. Consult the instructor BEFORE the absence to arrange for this accommodation. Line-of-duty absence and missed assignment policy: A student who is a member of the National Guard, Reserve, or other U.S. Armed Forces branch who misses classes, assignments or examinations due to line-of-duty responsibilities shall have the opportunity to make up the coursework in accordance with SSM 201-18 Accommodating Active Duty Military Personnel. This accommodation also applies to spouses who are the guardian of minor children during line-of -duty activities. This policy does not excuse students from course responsibilities during their absence. Students should first notify the Pat Tillman Veterans Center of their activation and then the instructor to discuss options.
Instructor Absence Policy	Students should wait for an absent instructor 15 minutes in class sessions of 90 minutes or less, and 30 minutes for those lasting more than 90 minutes, unless directed otherwise by someone from the academic unit.

Academic Integrity and Student Honor Code	The ASU student honor code affirms the commitment of ASU to uphold the values, principles, and ethics of academic integrity. Every student is expected to produce their original, independent work. Any student whose work indicates a violation of the ASU Academic Integrity Policy including cheating, plagiarism, and dishonesty will be subject to disciplinary action. Arizona State University and the Herberger Institute for Design and the Arts expect the highest standards of academic integrity from all students. Failure to meet these standards may result in suspension or expulsion from the university or other sanctions as specified in the ASU Student Academic Integrity Policy (http://provost.asu.edu/academicintegrity). Should you have any question about whether or not something falls subject to this clause, feel free to contact me or review the university policy on academic integrity at the above link or review: https://tutorials.lib.asu.edu/tutorials/rise/academic-integrity/index.html#/. Per ASU policy, a student may not avoid the consequences of academic dishonesty by withdrawing from a course, and may be placed back in the course in order to face sanctions resulting from academic integrity violations. You are responsible for abiding by this policy.
Student Learning Community Conduct and HIDA Professionalism Standard	ASU's Student Code of Conduct emphasizes the importance of intellectual, personal, social, and ethical development. It promotes a respectful and open environment for discourse, encouraging self-discipline and respect for others. Students are expected to treat instructors and peers with kindness and respect differing viewpoints in all communications, avoiding criticism or ridicule. Given the lack of nonverbal cues in electronic communication, students should also be mindful of how humor may be interpreted. Additionally, students must be aware of their Rights and Responsibilities at ASU. Further, Herberger Institute students are expected to adhere to the Herberger Institute Professionalism standards. Professionalism entails a set of skills critical for workplace and creative successes, and professional behavior creates an atmosphere promoting safe and high-quality spaces for constructive learning as well as individual and community well-being. The institute expects students to show professional behavior with clients, members of the community, and others in the university environment (e.g., classrooms, clinics, labs, studios) including members of the faculty and administration, other students and staff. Students are expected to conduct themselves in a professional manner, including arriving

	on time with an open attitude to learning, engaging in ethical behavior, resolving conflicts in an appropriate manner at all times, holding partnership information confidential, and using prudence in written and verbal communications. Professionalism includes: self-control, respectful communication, following all university and studio safety protocols and behavioral expectations, attendance and punctuality, honesty and integrity.
Disruptive, Threatening, or Violent Behavior	In the classroom and out, students are required to conduct themselves in a manner that promotes an environment that is safe and conducive to learning and conducting other university-related business. All incidents and allegations of violent or threatening conduct by an ASU student will be reported to the ASU Police Department (ASU PD) and the Office of the Dean of Students. Such incidents will be dealt with in accordance with the policies and procedures described in Section 104-02 of the Student Services Manual, available at https://public.powerdms.com/ASU/documents/1560490
Withdrawal	If you are unable to complete the course, it is your responsibility to arrange for withdrawal from the class. You will not be automatically withdrawn and unless you are officially withdrawn from the course you will receive a final grade based upon the total points you have earned for the semester. Students are required to pay all tuition and fees for any registered course unless enrollment is officially cancelled during the refund period. Please visit the Academic Calendar to review the withdrawal deadlines for this semester. For more information on Drop/Add and Withdrawal visit https://students.asu.edu/drop-add
Incomplete Policy	An incomplete is a limited academic exception (not a routine option) granted only when a student who is passing the course faces unexpected, serious circumstances beyond their control that prevent timely completion. It requires multiple stages of approval from academic leadership and is granted only for the minimum time needed to finish remaining work.
Special Accommodations	Your instructor is committed to making appropriate adjustments for any limitations caused by disabilities that have been documented with the Student Accessibility and Inclusive Learning Services (SAILS: https://eoss.asu.edu/drc). It is important to reach out to your instructor during office hours or by scheduling

	an appointment within the first week of class to discuss your accommodations and establish expectations. Please note that retroactive accommodations are seldom granted.
Title IX and Mandated Reporter Policy	Title IX is the federal law that provides that no person be excluded on the basis of sex from participation in, be denied benefits of, or be subjected to discrimination under any education program or activity. Both Title IX and university policy make clear that sexual violence and harassment based on sex is prohibited. Anyone who believes they have been subjected to sexual violence or harassed on the basis of sex can seek support, including counseling and academic support, from the university. If you or someone you know has been harassed on the basis of sex or sexually assaulted, you can find information and resources at https://sexualviolenceprevention.asu.edu/faqs. As a mandated reporter, I am obligated to report any information I become aware of regarding alleged acts of sexual discrimination, including sexual violence and dating violence. ASU Counseling Services, https://eoss.asu.edu/counseling, is available if you wish to discuss any concerns confidentially and privately. Arizona State University is committed to providing an environment free of discrimination, harassment, or retaliation for the entire university community, including all students, faculty members, staff employees, and guests. ASU expressly prohibits discrimination, harassment, and retaliation by employees, students, contractors, or agents of the university based on any protected status: race, color, religion, sex, national origin, age, disability, veteran status, sexual orientation, gender identity, and genetic information.
Copyright	Students may not upload or share any content that isn't their original work without complying with copyright laws. Faculty may remove content suspected of infringement. Additionally, all course materials including lectures, handouts, and notes are copyrighted and may not be shared, uploaded, sold, or distributed. See ACD 304–06 for more on “Commercial Note Taking Services.”

Artificial
Intelligence
Code of
Conduct

Arizona State University seeks to balance the promise of generative Artificial Intelligence (AI) tools to improve learning outcomes with the need for academic integrity, rigor and transparency.

With that in mind, sample syllabi statements for three distinct approaches are provided at the link below: (1) generative AI use generally permitted without course restrictions; (2) generative AI use permitted under some circumstances; and (3) no generative AI use permitted.

<https://docs.google.com/document/d/1u9RGD8oU3e16XkdFNC6k8tgXlzYW2L7yZ2ELW5LfZqQ/edit?usp=sharing>